

Alexandra Knigge

📍 Albuquerque, NM ✉ alex.m.knigge@gmail.com ☎ 505-850-7968 in alex-knigge 🌐 alexknigge

Skills

Programming: C++, C, Java, Python, Haskell, SQL

Parallel Computing: MPI, CUDA, OpenMP, OpenACC, UPC++

HPC Systems: Linux (Ubuntu, RHEL, Rocky), Warewulf, Slurm, Singularity, GNU Parallel

Education

University of New Mexico

Aug 2022 - Dec 2025

B.S. in Computer Science, Minor in Mathematics

GPA: 4.0/4.0, *summa cum laude*

- **Awards:** Goldwater Scholar, National Hispanic Scholar, Bob Veroff Awardee
- **Coursework:** High-Performance Computing, Parallel Processing, Data Structures and Algorithms, Machine Learning, Computer Architecture, Design of Large Programs, Ordinary Differential Equations

Experience

R&D Intern

May 2025 - Present

Sandia National Laboratories

- Designed and conducted a study on a monitoring framework, the Lightweight Distributed Metric Service (LDMS), to confirm it introduces negligible ($< 1\%$) overhead on application performance
- Developed containers with Slurm and Flux scheduler instances into a CI/CD pipeline to automate testing with virtual nodes and a SQL database for an open-source software
- Created a pipeline to automatically analyze user data across Sandia's HPC clusters and report on system errors on a daily basis

Undergraduate Research Assistant

June 2025 - Dec 2025

UNM Bienz Scalable Systems Laboratory

- Benchmarked Grace-Hopper nodes to analyze performance of power usage in CPU versus GPU processing
- Utilized NVIDIA's profiling tools such as NVIDIA-SMI and NSIGHT to gather performance data and automated collection through Bash scripts
- Wrote specialized GPU benchmarks using CUDA to test performance of hardware during intense workloads

HPC DevOps Specialist

June 2024 - May 2025

UNM Center for Advanced Research Computing

- Taught and developed multiple workshops covering HPC methods for computational fluid dynamics, quantum computing, and machine learning
- Optimized workflow pipelines using GNU parallel and slurm for climate and neurological science projects
- Utilized TensorFlow to optimize a machine learning classification model on an HPC cluster with GPUs

Undergraduate Research Assistant

Aug 2024 - May 2025

UNM Moses Biological Computation Laboratory

- Contributed to the development of a large-scale agent-based model that simulates fish foraging behavior on coral reefs
- Implemented parallelization with UPC++ to optimize agent interactions within the environment

Undergraduate R&D Intern

May 2023 - Aug 2024

Sandia National Laboratories

- Improved technology for space-based nuclear detonation detection systems through GUI development
- Developed cybersecurity software to assist in mitigating vulnerabilities present in code repositories
- Presented research findings to peers and colleagues, fostering discussion and feedback

Projects

UNM High-Performance Computing Team

Jan 2023 - Dec 2025

Team Manager

- Co-founded UNM's HPC team, coached and competed in the international SC23, SC24, and ISC24 Student Cluster Competitions
- Benchmarked clusters using applications such as HPL, HPCG, and MLPerf and then compiled and ran complex HPC applications on them such as ICON, WRF, and GROMACS
- Won first place in a Quantum-HPC competition which included a shot study and hardware analysis on IBM's quantum computers, summarized in a report

UNM Women in Computing Club

Aug 2024 - Dec 2025

President

- Developed outreach events to promote diversity and develop technical expertise in computer science
- Grew the club from five members to twenty members, including six new committee officers

Achievements

Bob Veroff Award, *University of New Mexico*

May 2025

Goldwater Scholar, *Barry Goldwater Scholarship & Excellence in Education Foundation*

March 2025

Brueckner Award, *Intersect360 Winter Classic Invitational*

April 2024

National Hispanic Scholar, *U.S. National College Board*

August 2022

Student of the Year, *Eldorado High School*

May 2022

Carson Scholar, *Carson Scholarship Foundation*

January 2021

Papers

A. Knigge, M. S. Swan, J. Green, Overhead Quantification of the Lightweight Distributed Metric Service for High Performance Computers. *WHPC Workshop at SC25*, St. Louis, MO, USA, November 17, 2025. [10.5281/zenodo.17554817](https://doi.org/10.5281/zenodo.17554817)


A. Knigge, M. S. Swan, J. Green, Comprehensive Overhead Study of the Lightweight Distributed Metric Service. *Technical Report*, The Computer Science Research Institute at Sandia National Laboratories, Albuquerque, NM and Livermore, USA. [SAND2025-142670](https://www.sandia.gov/publications/SAND2025-142670) 

Presentations

A. Knigge (10 June 2026) PID Data Tooling at Hyperscale. *LDMSCON2026*, Chicago, IL, USA

A. Knigge (29 July 2025) Containerizing Slurm, Developing an HPC Monitoring Pipeline, and Quantifying the Overhead of LDMS. *TITANS SW Symposium*, Albuquerque, NM, USA

A. Knigge, E. Wilcox, M. Burns (7 August 2024) Developing a New Software Suite for Nuclear Detection Simulation and Testing. *TITANS SW Symposium*, Albuquerque, NM, USA

A. Knigge, R. Iverson, E. Wilcox (28 June 2023) Mitigating Vulnerabilities for a Nuclear Detonation Detection Project. *TITANS SW Symposium*, Albuquerque, NM, USA

Posters

A. Knigge, M. Fricke (14 April 2025) Success of University-Industry Collaboration for HPC Education. *New Mexico Artificial Intelligence Consortium Poster Session*, Albuquerque, NM, USA

A. Knigge, M. Fricke (14 April 2025) Success of University-Industry Collaboration for HPC Education. *Center for Advanced Research Computing Annual Meeting*, Albuquerque, NM, USA

A. Knigge, K. A. Ahmed, M. Osinski (5 August 2022) Simulating Sidewall Roughness in Optical Waveguides. *UNM Center for High Technology Materials Poster Session*, Albuquerque, NM, USA

Teaching

High Performance Computing

Aug 2025 - Dec 2025

Teaching Assistant

- Gave lecture on Quantum Machine Learning on December 1st, 2025
- Taught and graded assignments for upper-level undergraduate/graduate HPC course
- Set up and configured 16-node cluster for students to use to learn about networking, operating systems, Slurm, and other HPC skills

Data Organization in C

Jan 2025 - May 2025

Teaching Assistant

- Developed and taught labs that covered memory and data organization topics in C
- Graded assignments and held office hours to help students